

GIOVANNI A. GOLLOTTI

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EDUCATION

City University of New York Graduate Center

Anticipated June 2027

M.S. in Astrophysics

Cumulative GPA: 4.0/4.0

University of Washington - Seattle

June 2025

B.S. in Physics & Astronomy (Honors in Astronomy)

Cumulative GPA: 3.56/4.0 (Astronomy GPA: 3.95/4.0)

RELEVANT RESEARCH EXPERIENCE

Data-Driven Identification of Candidate Nitrogen-Rich M Dwarfs

October 2025 - Present

Advisor: Dr. Carrie Filion

CUNY Graduate Center

- Sampling M dwarfs along a Gaia color-magnitude diagram with fitted MIST isochrones to locate metal-poor stars
- Applying the Nonnegative Matrix Factorization method to a synthetic M dwarf spectral grid in order to break spectra into its basic components
- Creating heat maps to determine how carbon, nitrogen, and oxygen affect each component across the spectral grid

Two Intriguing Transits of HIP 23309 Observed by TESS

June 2024 - March 2025

Advisors: Dr. Anastasios Tzanidakis, Dr. Tobin Wainer, & Prof. James Davenport *University of Washington*

- Analyzed the TESS photometric light curves and archival HARPS radial velocity measurements taken of the young, active M dwarf HIP 23309
- Obtained the epochs, duration, and period between two notable dips in order to determine the mass limits of a proposed sub-stellar companion
- Derived RV measurements from additional observations taken with the CHIRON spectrograph
- Applied a rejection sampling method to the sparse RV measurements to determine whether HIP 23309 is being gravitationally impacted by a sub-stellar companion

FIRST-AUTHOR PUBLICATIONS

1. **“Two Intriguing Transits of HIP 23309 Observed by TESS”**

Gollotti, G., et al. *RNAAS* 8, 293 (2024)

HIGHLIGHTED TALKS & PRESENTATIONS

Searching for Nitrogen Features in Low-Resolution BOSS M Dwarf Spectra

June 2026

Cool Stars 23 (Poster)

Tokyo, Japan

Two Distinct Transits of HIP 23309 Observed by TESS

January 2025

AAS 245 Meeting (Poster)

National Harbor, MD

HIP 23309's Mysterious Light Curve Variability

September 2024

DiRAC Summer Prize Winner Presentations (Talk)

Seattle, WA

Early LSST Template Generation Strategies

August 2023

Rubin 2023 Project & Community Workshop Conference (Poster)

Tucson, AZ

HONORS & AWARDS

Astronomy Honors	Spring 2025
Chambliss Student Award Winner	January 2025, 245th Meeting of the AAS
DiRAC Summer Research Prize Winner	\$3,500, Summer 2024
Dean's List	Wi/Su/Au 2023, Wi/Sp/Au 2024, & Wi/Sp 2025

EMPLOYMENT

Post-Baccalaureate Research Assistant (Remote)	May - July 2025
<i>Department of Physics & Astronomy</i>	<i>Northwestern University</i>

- Created automated gridding method that generates cutouts of PHAT-survey images
- Deleted cutouts without data or did not include any stellar clusters
- Sorted stellar cluster catalogs to isolate clusters associated with specific PHAT cutouts

Teaching Assistant	July 2024
<i>UW Robinson Center</i>	<i>University of Washington</i>

- **Class:** From Galileo to Einstein and Beyond
- Collaborated with instructors and fellow TA with creating lesson plans, labs, and grading rubrics
- Demonstrated flexibility and adaptability in the classroom to meet the changing needs of my instructors and students
- Graded middle and high school students' work in a timely manner while giving constructive feedback

Student Grader	Spring/Autumn 2024 & Winter/Spring 2025
<i>UW Department of Astronomy</i>	<i>University of Washington</i>

- **Classes:** ASTR 101: Astronomy & ASTR 150: The Planets
- Assisted with asynchronous grading, note-keeping, and communication between professor and students
- Collaborated with professor and graders in order to create assignments that provoke thoughtful discussions
- Facilitated meaningful online discussions between students in various class sections

LEADERSHIP

CUNY Astro Student Union	September 2025 - Present
<i>Student Representative</i>	<i>CUNY Graduate Center</i>

- Act as the Masters program representative during department meetings with faculty
- Collaborate with other student representatives to create monthly newsletters, social events, and workshops

Sexual-Orientation and Gender Minorities in Astronomy at UW	September 2023 - June 2025
<i>Co-Chair</i>	<i>University of Washington</i>

- Collaborated with other clubs to create connections between students in both the Physics & Astronomy Departments
- Connected undergraduates with graduates, postdocs, and faculty offering new research opportunities
- Facilitated meetings with students and visiting colloquium speakers through weekly Tea Talks
- Strengthened community of LGBTQ+ astronomers through hosting interactive social events

TECHNICAL SKILLS

Computer Languages	Python, Bash, Java
Astronomy Software	Astropy, Lightkurve, DS9, Emcee, Korg, Scikit-learn
Data Analysis	Photometry & Spectroscopy, Convolution, Data Reduction
Other Software	LaTeX, Github, Overleaf, Topcat